



Department of Information Technology
Academic Year 2024 – 2025(Even Semester)

Degree, Semester & Branch: B.Tech-VI & IT

Course Code & Title: CCS356 & Object Oriented Software Engineering

Name of the Faculty member (s): M.Rethina kumari

Innovative Practice Description

- **Unit / Topic: Unit II / Formal System Specification**

Course Outcome: The student will be able to understand the fundamental principles of **formal system specification**, which involves the precise mathematical modeling of software systems.

- **Activity Chosen: One Minute Paper**

- **Justification:**

In the one-minute paper activity, students will be given five minutes to write a brief summary of the class session. Formal system specification is a fundamental concept in software engineering, involving precise mathematical descriptions of software systems. To reinforce understanding and recall of formal specification techniques, this activity is chosen.

- **Time Allotted for the Activity: 5 Minutes**

- **Details of the Implementation:**

- ✓ The faculty explained the concept of formal system specification, including its importance and methodologies.
- ✓ Students were asked to summarize key points of formal system specification on paper within one minute.
- ✓ The faculty collected and reviewed the responses. Students who made errors were given feedback and asked to rewrite their responses.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PSO1	PSO3
CO1	3	3	2	3	1	2	2

(1 – Low

2 – Moderate

3 – High)



- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO9	PSO1	PSO3
Justification for correlation	Students will be able to recall and understand formal system specification, which is essential for software engineering .	Students will apply the learned concepts in system modeling and analysis ..	Students will develop solutions for software reliability using formal methods.	Individual involvement improves critical thinking and self-assessment .	Emphasize continuous learning by keeping up with the latest advancements in tools, frameworks, and methodologies .	Develop precise and accurate software requirements specification using UML models and formal analysis methods. Ensure compliance with technical standards.	Combine functional and non-functional requirements to create reliable, scalable systems. Ensure the software system meets quality and usability standards.



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- Images / Screenshot of the practice:



Fig 1: Time given to prepare on the topic



Fig 2: Students are on the task

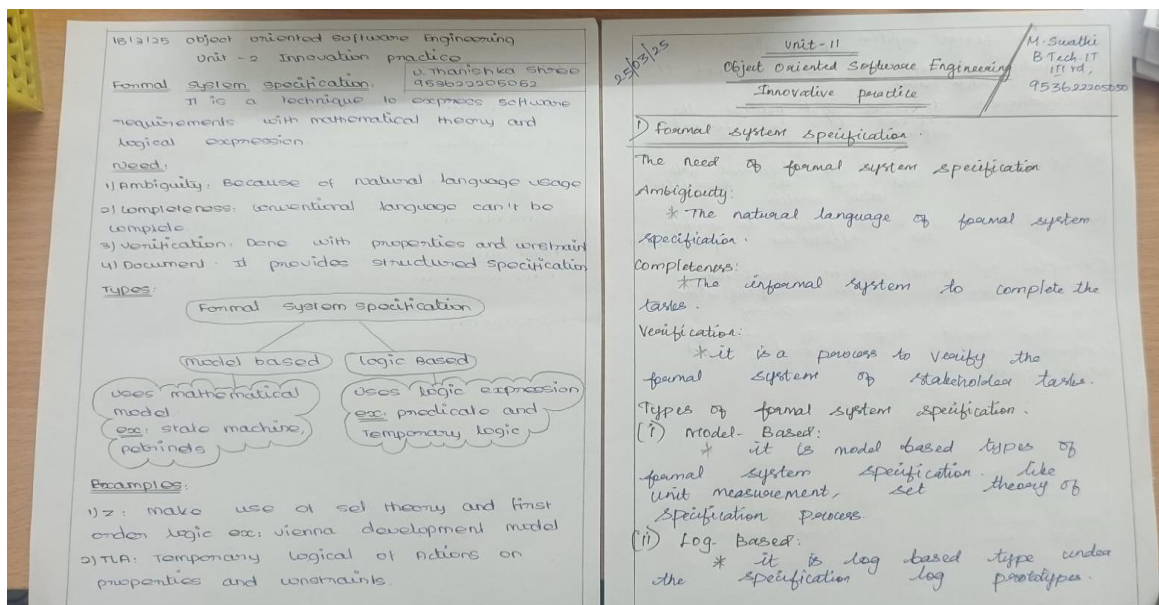


Fig 3: Proof of the task: U.Thanishka Shree and M.Swathi

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- ✓ Students reported that summarizing key concepts helped reinforce their understanding of formal system specification.
- ✓ Students felt the activity would be beneficial for future software engineering courses.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Improved retention and understanding of formal system specification.
- ✓ Identification of knowledge gaps to address in future sessions.

❖ Challenges faced in implementation:

- ✓ Some students initially struggled with summarizing technical concepts concisely.
- ✓ A few students required additional time to complete the activity.

References:

- ✓ <https://www.slideshare.net/formalspecification>
- ✓ <https://www.academicpapers.org/formalspecification>

Signature of Faculty Member

HOD